

# AUTUMN SEMESTER EXAMINATION-2023

Course Code: PHYUGBS01

Course Title: Engineering Physics

Department of Appearing Students: CEN, EEN, MEN, ECE, CSE, 1<sup>st</sup> Year, SEM-I

Full Marks: 80

Time: 3 hrs

Answer any EIGHT (08) questions

10 X 8 = 80

1. (i) Find the angle between  $\mathbf{A}=2\mathbf{i}+2\mathbf{j}-\mathbf{k}$  and  $\mathbf{B}=6\mathbf{i}-3\mathbf{j}+2\mathbf{k}$  ✓  
 (ii) Determine the value of  $a$  so that  $\mathbf{A}=2\mathbf{i}+a\mathbf{j}+\mathbf{k}$  and  $\mathbf{B}=4\mathbf{i}-2\mathbf{j}-2\mathbf{k}$  are perpendicular.  
 (iii) Find the projection of the vector  $\mathbf{A}=\mathbf{i}-2\mathbf{j}+\mathbf{k}$  on the vector  $\mathbf{B}=4\mathbf{i}-4\mathbf{j}+7\mathbf{k}$   
 (iv) Find out a unit vector perpendicular to the two vectors  $-\mathbf{i}+\mathbf{j}+\mathbf{j}$  and  $2\mathbf{i}+3\mathbf{j}+2\mathbf{k}$ . ✗  
 (v) On application of a force  $7\mathbf{i}+2\mathbf{j}-\mathbf{k}$  N, a particle moves from position  $-4\mathbf{i}+3\mathbf{j}+6\mathbf{k}$  m to another position  $2\mathbf{i}-6\mathbf{j}+5\mathbf{k}$ . Calculate the work done. ✓

2x5=10

2. (i) Define Torque.  
 (ii) Prove that for a system of particles  $\mathbf{L}=\mathbf{I}\boldsymbol{\omega}$ , symbols are usual meanings.  
 (iii) Define Radius of Gyration.  
 (iv) A fly wheel of mass 500 kg and diameter 2 m is revolving at the speed of 500 rpm. If whole mass of the wheel is concentrated at the edge, calculate its (a) angular velocity, (b) moment of inertia, (c) energy.

2+3+2+3=10

3. (i) Define Viscosity of liquid.  
 (ii) What is Critical Velocity?  
 (iii) State and prove Stoke's law.

2+2+6=10

4. (i) State Coulomb's law in electrostatics.  
 (ii) Define electric line of force and write its properties.  
 (iii) State and prove Gauss's theorem in electrostatics.

2+(2+2)+4=10

5. (i) Calculate the Electric field due to a uniformly charged sphere using the Gauss's law.  
 (ii) Define Electric dipole.  
 (iii) Calculate the electric field in free space due to electric dipole at axial point.

4+2+4=10

6. (i) Derive the expression for the fringe width in a Young's double slit experiment.  
 (ii) Define 'Fringes of equal thickness' (FET).  
 (iii) What is the difference between Fraunhofer diffraction and Fresnel diffraction?

5+2+3=10

OR

- (i) What is interference of light? Using the theory of superposition of wave, derive the condition of constructive and destructive interference.  
 (ii) Write Huygens principle of wave optics. What are the secondary wavelets

(3+6)+(2+2)=10

7. (i) Define H-polaroid.  
 (ii) An EM waves  $E_x(z, t) = E_{0x} \cos(kz - \omega t)$  and  $E_y(z, t) = E_{0y} \cos(kz - \omega t + \delta)$ . Find the state of polarization in following cases (a)  $\delta = 0, E_{0x} \neq E_{0y}$ ; (b)  $\delta = \frac{\pi}{2}, E_{0x} = E_{0y}$ ;  
 (iii) How would you combine two linear vibrations to get circular vibration?  
 (iv) A plane polarized light become circularly polarized light by passing through a crystal of thickness  $3 \times 10^{-3} \text{ cm}$ . Calculate the difference in refractive indices for E ray and O ray if  $\lambda = 6000 \text{ \AA}$ .  
 $2 + (2+2) + 2 + 2 = 10$

OR

- (i) Derive the expression of intensity  $I = I_0 \frac{\sin^2 \alpha}{\alpha^2}$  in single slit diffraction, where  $\alpha = \frac{\pi a}{\lambda} \sin \theta$ .  
 (ii) Using the formula derive the position of primary minima of the diffraction pattern  
 $5 + 5 = 10$

- ✓ 8. (i) What is Einstein's A, B co-efficient? Derive a relation between them.  
 (ii) Discuss what the basic difference between spontaneous and stimulated emission are.  
 (iii) How population inversion is obtained in practical lasers?  
 $(2+4) + 2 + 2 = 10$

OR

- (i) A monochromatic light of 300 nm is used in Young's double slit experiment. If the screen is placed at 100cm and the distance between two slits are 1 mm then find the fringe width, position 1<sup>st</sup> bright fringe.  
 (ii) A monochromatic light of 300 nm is used in single slit diffraction. If the slit width is 0.1 mm, then find the width of central diffraction maxima  
 (iii) Draw the combined pattern of diffraction and interference in Young's double slit experiment.  
 $4 + 4 + 2 = 10$

- ✓ 9. (i) What is Mass defect of a nucleus? A Copper Nucleus has a mass number 64 and atomic number 29. The neutron and the proton mass are given as 1.0087 amu and 1.0073 amu. The mass of the copper nucleus is 63.546u. Calculate the mass defect and Binding energy of the Copper Nucleus. Given  $1 \text{ u} = 931.5 \text{ MeV}$ .  
 (ii) Explain the Volume effect, Surface effect and Coulomb effect in Bethe Weizsäcker formula of semi empirical mass.  
 $(2+3) + 5 = 10$

- ✓ 10. (i) Write down the properties of nuclear force?  
 (ii) Define atomic number and mass number of an atom.  
 (iii) The mass number of an atom is 40. Find the nuclear radius?  
 $4 + 2 + 3 = 10$
- ✓ 11. (i) Show that, the number of undecayed radioactive nuclei (N) of a sample at time (t) is  $N(t) = N_0 e^{-\lambda t}$ , where,  $N_0$  is the number of undecayed nuclei at a time  $t = 0$  and  $\lambda$  is decay constant.  
 (ii) In a radioactive sample number of undecayed nuclei decreases from 16000 to 1000 in 12 days. Find the half-life and decay constant.  
 (iii) Define isotope with an example.  
 $5 + 4 + 1 = 10$

8. Evaluate

$$\iint dxdy$$

over the region bounded by  $y = x^2$  and  $x = y^2$ .

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9. Evaluate

$$\iint (x^2 + y^2) dxdy$$

over the region bounded by  $xy = 1$ ,  $y = 0$ ,  $y = x$  and  $x = 2$ .

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10. Evaluate

$$\iiint (x + y + z + 1) dxdydz$$

over the region defined by  $x \geq 0$ ,  $y \geq 0$ ,  $z \geq 0$ ,  $x + y + z \leq 1$ .

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11. Use Green's Theorem to evaluate

$$\oint_C (xydx + x^2y^3dy)$$

where  $C$  is the triangle with vertices  $(0, 0)$ ,  $(1, 0)$ ,  $(1, 2)$  with positive orientation.

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12. Use Divergence Theorem to evaluate

$$\iint_S \vec{F} \cdot d\vec{S}$$

where  $\vec{F} = \sin(\pi x)\hat{i} + xy^3\hat{j} + (z^2 + 4x)\hat{k}$  and  $S$  is the surface of the box with  $-1 \leq x \leq 2$ ,  $0 \leq y \leq 1$  and  $1 \leq z \leq 4$ .

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